78 Lumbar Disk Arthroplasty

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INDICATIONS

- Lumbar disk arthroplasty (LDA) is indicated as a treatment of chronic, incapacitating low back pain that is diskogenic in origin at the L4-5 or L5-S1 level and not accompanied by neural element compression resulting in claudication or radiculopathy. Diagnosis should be documented with magnetic resonance imaging (MRI), plain lumbar spine x-rays, and positive results of provocative diskography of the pathologic level.
- To be considered for surgery, patients should have failed at least 6 months of conservative nonsurgical therapies and ideally be 18 to 50 years old. Patients who have previously undergone posterior disk interventions, such as diskectomy or nucleolysis, may be candidates for LDA as long as no acute neural compression is present and the remaining facet anatomy is sufficient to prevent distraction of the disk space and provide stability of the segment to be operated.

CONTRAINDICATIONS

- Assuming a patient's general medical condition is adequate to undergo elective spine surgery, contraindications to this procedure are active diskitis, previous (failed) fusion surgery at the pathologic level, malignancy, fracture or spondylolysis of the adjacent vertebrae, spondylolisthesis of the pathologic segment, osteopenia insufficient to support the disk prosthesis, and advanced facet arthrosis.
- As with other anterior lumbar spine procedures, relative contraindications to this approach include the presence of an infrarenal aortic aneurysm, congenital or iatrogenic genitourinary anatomic abnormalities such as an ipsilateral single ureter or kidney, or a history of previous retroperitoneal surgery.

PLANNING AND POSITIONING

Fig. 78.1 The patient is positioned supine on the operating table. The lumbar spine is placed in extension; the patient's legs are abducted if a lithotomy position is used. The surgeon performs the procedure standing between the legs of the patient if the lithotomy position is used. This position gives the surgeon a slightly more centered approach during prosthesis placement, which can be crucial to the success of the device.

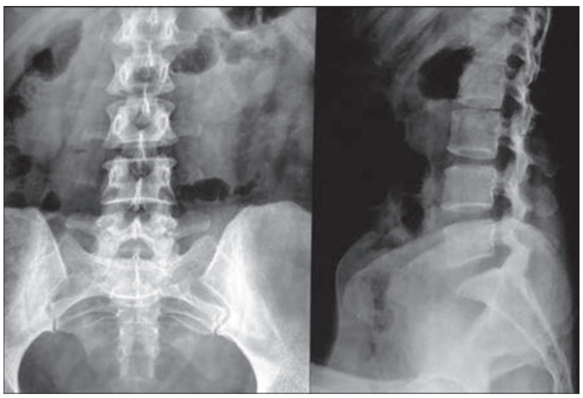


Fig. 78.2 Before incision, the correct disk space is localized using anteroposterior and lateral fluoroscopy, and the skin is marked appropriately. The incision for this approach corridor is centered at this location and marked.

PROCEDURE

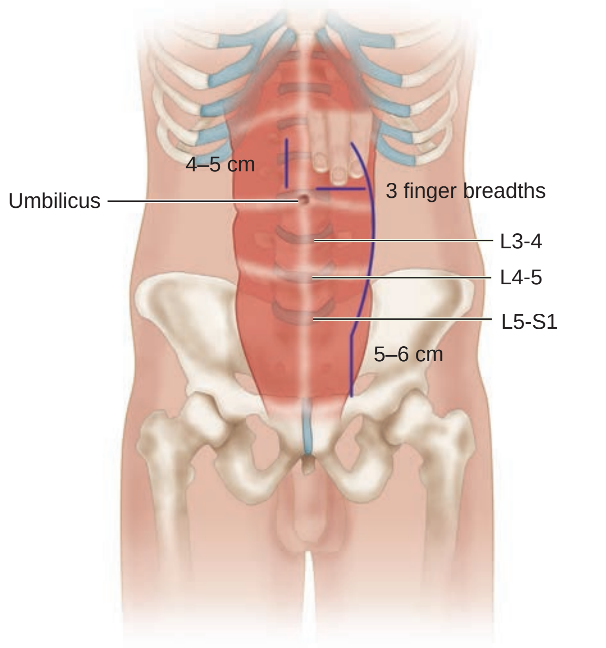


Fig. 78.3 Transperitoneal or retroperitoneal approaches may be used to access the L4-5 and L5-S1 disk spaces. Both approaches may be performed via various incisions, including midline, paramedian, and Pfannenstiel incisions. An approach from the left side is generally performed because gentle manual retraction of the aorta is more safely performed than retraction of the inferior vena cava. Most surgeons prefer to use the retroperitoneal mini-open approach because of lower rates of hollow viscus injury, retrograde ejaculation, and postoperative ileus. Final fluoroscopic confirmation of the correct disk level is mandatory when the anterior spine is visualized intraoperatively. The anatomic midline is determined with anteroposterior fluoroscopy and marked with a screw above the disk to be removed.

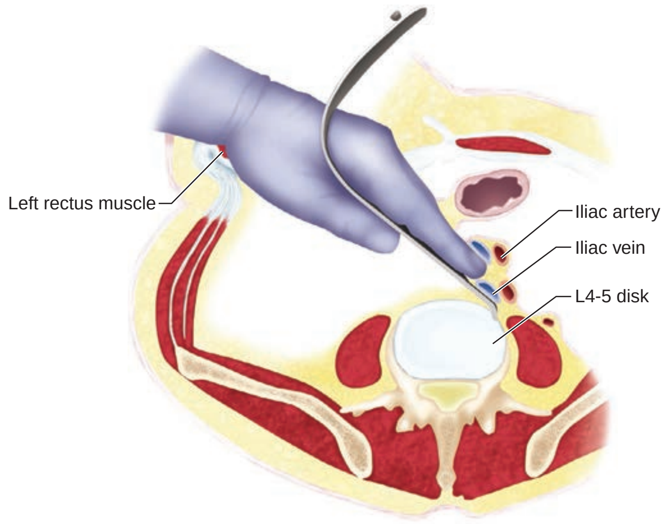


Fig. 78.5 At the L4-5 level, mobilization of the left iliac vessels from left to right is performed after identification and ligation of the iliolumbar vein. This vein enters the common iliac at this level, and avulsion at this anastomosis can lead to aggressive, unnecessary bleeding. Disk replacement devices typically require a large footprint for proper biomechanical function, contact with the outer bony vertebral rim (where bone is stronger), and prevention of subsidence. Lateral vessel retraction is frequently necessary. This retraction can be achieved manually using vein retractors or with Hohmann retractors anchored into the vertebrae. Table-mounted retractors allow the surgeon to work without an assistant, and Kirschner wires drilled into the vertebral bodies can be used for gentle vessel retraction where more forceful retraction of retroperitoneal structures is unnecessary.

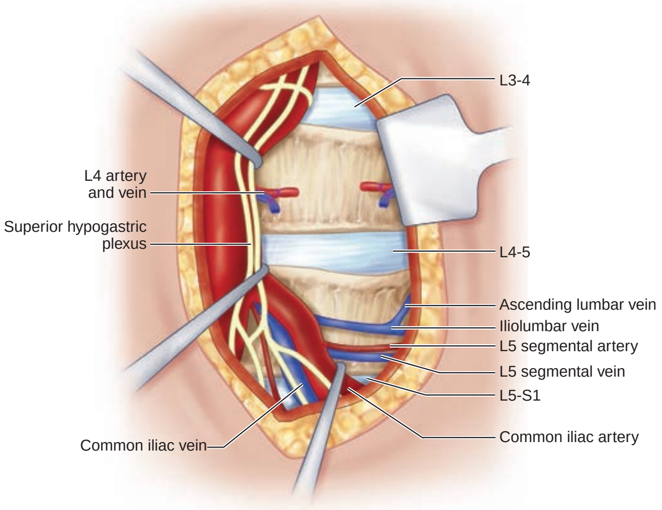


Fig. 78.4 Prevertebral tissue at the L5-S1 level is opened using a vertical incision. Bipolar electrocautery is used exclusively; monopolar electrocautery is avoided at this level in an effort to spare injury to the autonomic nerves traversing this space. If these nerves are injured, impotence and infertility secondary to retrograde ejaculation in men may result. Mobilization of the large vessels is not commonly necessary at this level.

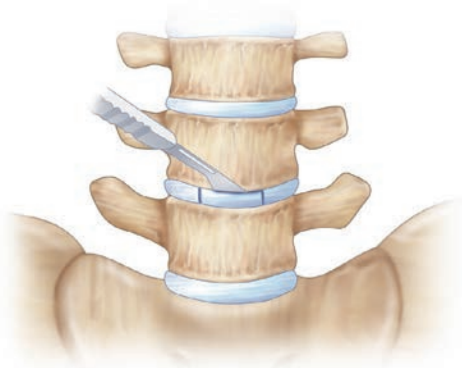
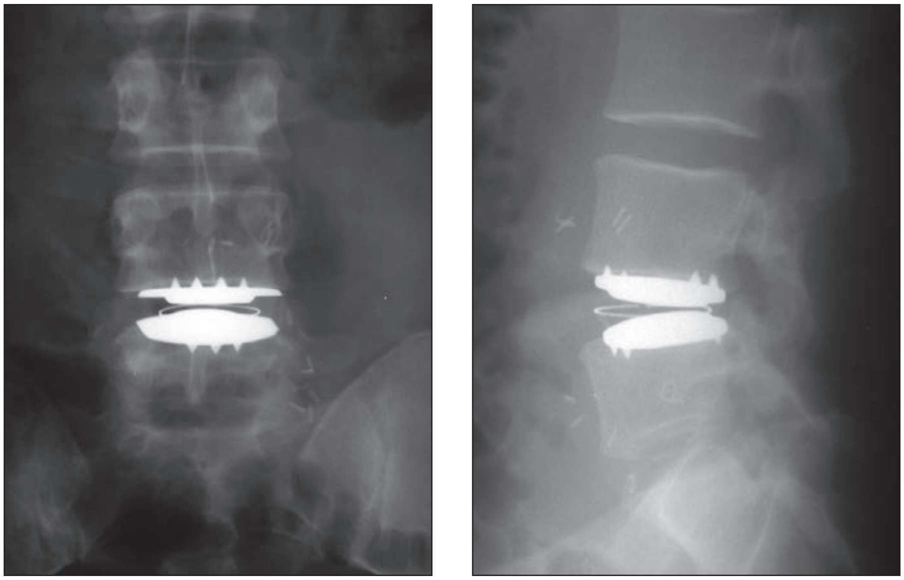


Fig. 78.6 Incision of the anterior annulus of the disk is performed, taking care to leave the lateral annular walls intact. The nucleus pulposus and all disk tissues except for the lateral annular walls are completely removed. Both vertebral end plates are left intact. Careful attention is needed to ensure adequate removal and release of the posterior disk to allow parallel intervertebral distraction. The intervertebral disk height is restored by distracting the space with a central spreader and distracting chisels of graduated sizes.

Fig. 78.7 Optimal coverage of the cross-sectional area of the vertebral end plates is performed by placing various sizing templates within the disk space and confirming the appropriate fit with intraoperative fluoroscopy. Marking the center of the disk space can be useful to ensure proper centering of the implant because its position determines the instantaneous axis of rotation of that spinal segment.



A B

Fig. 78.8 The disk prosthesis is put into place. Most devices require either the use of a specialized chisel to crease end plate troughs to accommodate a stabilizing keel or teeth on the total disk replacement end plates. Careful attention is paid to positioning within the disk space. In the coronal plane **(A,** anteroposterior fluoroscopy), it is centered in the midline, but in the sagittal plane **(B,** lateral fluoroscopy), it is centered 2 mm posterior to midline. This location recapitulates the physiologic instantaneous axis of rotation throughout the flexion-extension arc of the normal disk as documented by Gertzbein and colleagues. Hemostasis is achieved, and the incision is closed in a usual layered fashion.

Fig. 78.9 Photograph of LDA seen in Fig. 78.8 in place in the patient.

TIPS FROM THE MASTERS

- Exposure of the lateral margins of the disk space is greater than that needed for anterior lumbar interbody fusion. It is useful to work with a vascular or exposure surgeon who can obtain this lateral dissection of the major vessels safely.
- Consent should be obtained in every case for the possibility of intraoperative conversion to a fusion procedure if prosthesis fit or bone quality is found to be suboptimal.
- A bowel preparation should be employed before surgery to make exposure easier and reduce constipation postoperatively.
- The largest prosthesis that can be accommodated should be used to improve disk mechanics and reduce the risk of subsidence.
- Hybrid constructs (L5-S1 anterior lumbar interbody fusion and L4-5 total disk replacement) can be used to treat two-level disease if it is undesirable to perform a two-level LDA.

PITFALLS

Avoid overdistraction of the disk space because this can lead to postoperative sciatic pain.

In patients in whom a previous laminotomy or microdiskectomy has been performed at the index level, exacerbation of sciatic pain can occur during disk mobilization, but this is typically transient.

Patients with facet joint arthrosis should not undergo LDA. Arthrosis can be identified by high T2 signal on axial MRI, test injections into the joints, or single photon emission computed tomography nuclear medicine scans.

BAILOUT OPTIONS

- In rare cases where placement of the disk prosthesis is impossible after disk removal, either anterior or posterior interbody fusion may be necessary.